

Koryak Fine-Grained Maximal Pairwise brms Models: Progress Summary

Generated 2026-05-19 10:00 from saved outputs in output/koryak_brms_fine_grained_pairwise_maximal. Completed model count summarized here: 18 gaze models.

Run Status

Completed contrast folders: 9/10. **Partial/running:** 1. **Not started:** 0. A contrast folder is treated as complete only when both agent and patient fixed-effect and diagnostic CSVs are present.

Window	Comparison	Status	Detail
Linguistic encoding 1	Direct 1-1 animate patient vs direct 1-2 animate patient	complete	Agent and patient summaries present
Linguistic encoding 1	Direct 1-1 inanimate patient vs direct 1-2 inanimate patient	complete	Agent and patient summaries present
Linguistic encoding 1	Inverse 2-2 animate patient vs inverse 2-1 animate patient	complete	Agent and patient summaries present
Linguistic encoding 1	Direct 1-1 animate patient vs inverse 2-2 animate patient	complete	Agent and patient summaries present
Linguistic encoding 1	Direct 1-1 inanimate patient vs inverse 2-2 inanimate patient	complete	Agent and patient summaries present
Linguistic encoding 3	Direct 1-1 animate patient vs direct 1-2 animate patient	complete	Agent and patient summaries present
Linguistic encoding 3	Direct 1-1 inanimate patient vs direct 1-2 inanimate patient	complete	Agent and patient summaries present
Linguistic encoding 3	Inverse 2-2 animate patient vs inverse 2-1 animate patient	complete	Agent and patient summaries present
Linguistic encoding 3	Direct 1-1 animate patient vs inverse 2-2 animate patient	complete	Agent and patient summaries present
Linguistic encoding 3	Direct 1-1 inanimate patient vs inverse 2-2 inanimate patient	partial/running	Missing fixed, diagnostics, all_diagnostics, sampler

Model Specification

Outcome is empirical-logit fixation proportion, fit separately for agent looks and patient looks. Time is modeled up to cubic using orthogonal polynomials. The current run uses the maximal correlated random-effect structure requested for participants and lexical items.

Formula: logit(looks) ~ poly(time, 3) * contrast + (1 + poly(time, 3) * contrast | participant) + (1 + poly(time, 3) | participant_condition) + (1 + poly(time, 3) * contrast | item)

Priors: no explicit priors are set in the script, so brms default weakly informative priors are used.

Chains	Iter	Warmup	adapt_delta	max_treedepth	file_refit
4	2000	1000	0.99	12	on_change

Included Data

Window	Comparison	Condition	Sentences/trials	Participants	Lexical items	Time bins
Linguistic encoding 1	Direct 1-1 animate patient vs direct 1-2 animate patient	direct 1 agent 1 patient animate patient	99	16	14	2047
Linguistic encoding 1	Direct 1-1 animate patient vs direct 1-2 animate patient	direct 1 agent 2 patients animate patient	106	16	15	2232
Linguistic encoding 1	Direct 1-1 inanimate patient vs direct 1-2 inanimate patient	direct 1 agent 1 patient inanimate patient	95	16	13	1860
Linguistic encoding 1	Direct 1-1 inanimate patient vs direct 1-2 inanimate patient	direct 1 agent 2 patients inanimate patient	92	15	15	1894
Linguistic encoding 1	Inverse 2-2 animate patient vs inverse 2-1 animate patient	inverse 2 agents 1 patient animate patient	91	15	15	2102
Linguistic encoding 1	Inverse 2-2 animate patient vs inverse 2-1 animate patient	inverse 2 agents 2 patients animate patient	89	16	16	2019
Linguistic encoding 1	Direct 1-1 animate patient vs inverse 2-2 animate patient	direct 1 agent 1 patient animate patient	99	16	14	2047
Linguistic encoding 1	Direct 1-1 animate patient vs inverse 2-2 animate patient	inverse 2 agents 2 patients animate patient	89	16	16	2019
Linguistic encoding 1	Direct 1-1 inanimate patient vs inverse 2-2 inanimate patient	direct 1 agent 1 patient inanimate patient	95	16	13	1860
Linguistic encoding 1	Direct 1-1 inanimate patient vs inverse 2-2 inanimate patient	inverse 2 agents 2 patients inanimate patient	74	14	15	1357

Window	Comparison	Condition	Sentences/trials	Participants	Lexical items	Time bins
Linguistic encoding 3	Direct 1-1 animate patient vs direct 1-2 animate patient	direct 1 agent 1 patient animate patient	99	16	14	2079
Linguistic encoding 3	Direct 1-1 animate patient vs direct 1-2 animate patient	direct 1 agent 2 patients animate patient	106	16	15	2226
Linguistic encoding 3	Direct 1-1 inanimate patient vs direct 1-2 inanimate patient	direct 1 agent 1 patient inanimate patient	95	16	13	1995
Linguistic encoding 3	Direct 1-1 inanimate patient vs direct 1-2 inanimate patient	direct 1 agent 2 patients inanimate patient	92	15	15	1932
Linguistic encoding 3	Inverse 2-2 animate patient vs inverse 2-1 animate patient	inverse 2 agents 1 patient animate patient	91	15	15	1911
Linguistic encoding 3	Inverse 2-2 animate patient vs inverse 2-1 animate patient	inverse 2 agents 2 patients animate patient	89	16	16	1869
Linguistic encoding 3	Direct 1-1 animate patient vs inverse 2-2 animate patient	direct 1 agent 1 patient animate patient	99	16	14	2079
Linguistic encoding 3	Direct 1-1 animate patient vs inverse 2-2 animate patient	inverse 2 agents 2 patients animate patient	89	16	16	1869

Contrast Effects with 95% CI Excluding Zero

Window	Comparison	Looks	Term	Direction	Beta	95% CI	P(beta > 0)
Linguistic encoding 1	Direct 1-1 inanimate patient vs inverse 2-2 inanimate patient	agent looks	Time 1 x contrast	negative	-19.554	[-39.595, -0.347]	0.024

Fixed Effects: Contrast Terms

Contrasts use +1/-1 coding: the reported beta is half of the full condition difference. The full-difference column is 2*beta. Green rows have a 95% credible interval excluding zero.

Linguistic encoding 1

Comparison	Looks	Term	Beta	95% CI	P(beta > 0)	Full difference
Direct 1-1 animate patient vs direct 1-2 animate patient	agent looks	Main contrast	-0.056	[-0.495, 0.419]	0.391	-0.112 [-0.989, 0.838]
Direct 1-1 animate patient vs direct 1-2 animate patient	agent looks	Time 1 x contrast	0.990	[-15.656, 17.555]	0.544	1.981 [-31.312, 35.110]
Direct 1-1 animate patient vs direct 1-2 animate patient	agent looks	Time 2 x contrast	-0.403	[-13.652, 12.800]	0.476	-0.806 [-27.305, 25.600]
Direct 1-1 animate patient vs direct 1-2 animate patient	agent looks	Time 3 x contrast	8.124	[-4.136, 20.325]	0.906	16.247 [-8.271, 40.650]
Direct 1-1 animate patient vs direct 1-2 animate patient	patient looks	Main contrast	-0.272	[-0.846, 0.305]	0.174	-0.543 [-1.693, 0.611]
Direct 1-1 animate patient vs direct 1-2 animate patient	patient looks	Time 1 x contrast	-1.807	[-22.644, 20.169]	0.429	-3.615 [-45.288, 40.338]
Direct 1-1 animate patient vs direct 1-2 animate patient	patient looks	Time 2 x contrast	10.011	[-11.626, 30.781]	0.834	20.022 [-23.252, 61.562]
Direct 1-1 animate patient vs direct 1-2 animate patient	patient looks	Time 3 x contrast	-9.949	[-23.243, 3.537]	0.068	-19.898 [-46.486, 7.075]
Direct 1-1 inanimate patient vs direct 1-2 inanimate patient	agent looks	Main contrast	0.211	[-0.084, 0.524]	0.922	0.421 [-0.167, 1.047]
Direct 1-1 inanimate patient vs direct 1-2 inanimate patient	agent looks	Time 1 x contrast	-4.543	[-24.135, 14.730]	0.314	-9.086 [-48.269, 29.459]
Direct 1-1 inanimate patient vs direct 1-2 inanimate patient	agent looks	Time 2 x contrast	-4.962	[-23.751, 14.633]	0.298	-9.924 [-47.502, 29.266]
Direct 1-1 inanimate patient vs direct 1-2 inanimate patient	agent looks	Time 3 x contrast	-5.710	[-18.909, 8.870]	0.192	-11.419 [-37.818, 17.740]
Direct 1-1 inanimate patient vs direct 1-2 inanimate patient	patient looks	Main contrast	-0.116	[-0.494, 0.241]	0.270	-0.233 [-0.988, 0.481]
Direct 1-1 inanimate patient vs direct 1-2 inanimate patient	patient looks	Time 1 x contrast	-3.058	[-29.202, 23.956]	0.412	-6.116 [-58.404, 47.913]
Direct 1-1 inanimate patient vs direct 1-2 inanimate patient	patient looks	Time 2 x contrast	7.442	[-15.212, 29.713]	0.748	14.884 [-30.424, 59.426]
Direct 1-1 inanimate patient vs direct 1-2 inanimate patient	patient looks	Time 3 x contrast	-3.457	[-20.184, 12.775]	0.342	-6.913 [-40.368, 25.550]
Inverse 2-2 animate patient vs inverse 2-1 animate patient	agent looks	Main contrast	0.030	[-0.354, 0.414]	0.557	0.059 [-0.709, 0.828]
Inverse 2-2 animate patient vs inverse 2-1 animate patient	agent looks	Time 1 x contrast	9.365	[-13.708, 33.266]	0.778	18.729 [-27.416, 66.533]
Inverse 2-2 animate patient vs inverse 2-1 animate patient	agent looks	Time 2 x contrast	0.203	[-28.451, 29.004]	0.511	0.406 [-56.901, 58.009]
Inverse 2-2 animate patient vs inverse 2-1 animate patient	agent looks	Time 3 x contrast	4.271	[-6.917, 16.088]	0.773	8.542 [-13.835, 32.176]
Inverse 2-2 animate patient vs inverse 2-1 animate patient	patient looks	Main contrast	-0.135	[-0.687, 0.399]	0.313	-0.271 [-1.375, 0.797]
Inverse 2-2 animate patient vs inverse 2-1 animate patient	patient looks	Time 1 x contrast	-12.598	[-39.848, 14.123]	0.179	-25.197 [-79.696, 28.245]

Comparison	Looks	Term	Beta	95% CI	P(beta > 0)	Full difference
Inverse 2-2 animate patient vs inverse 2-1 animate patient	patient looks	Time 2 x contrast	19.977	[-4.566, 44.685]	0.950	39.954 [-9.133, 89.369]
Inverse 2-2 animate patient vs inverse 2-1 animate patient	patient looks	Time 3 x contrast	4.045	[-9.899, 17.524]	0.726	8.090 [-19.798, 35.049]
Direct 1-1 animate patient vs inverse 2-2 animate patient	agent looks	Main contrast	-0.016	[-0.294, 0.258]	0.452	-0.033 [-0.588, 0.517]
Direct 1-1 animate patient vs inverse 2-2 animate patient	agent looks	Time 1 x contrast	10.837	[-8.956, 30.455]	0.854	21.675 [-17.911, 60.910]
Direct 1-1 animate patient vs inverse 2-2 animate patient	agent looks	Time 2 x contrast	4.211	[-16.342, 25.376]	0.660	8.423 [-32.684, 50.753]
Direct 1-1 animate patient vs inverse 2-2 animate patient	agent looks	Time 3 x contrast	-1.582	[-11.655, 8.331]	0.383	-3.164 [-23.311, 16.662]
Direct 1-1 animate patient vs inverse 2-2 animate patient	patient looks	Main contrast	-0.199	[-0.611, 0.208]	0.162	-0.398 [-1.221, 0.417]
Direct 1-1 animate patient vs inverse 2-2 animate patient	patient looks	Time 1 x contrast	-7.907	[-32.602, 18.124]	0.267	-15.814 [-65.204, 36.247]
Direct 1-1 animate patient vs inverse 2-2 animate patient	patient looks	Time 2 x contrast	-8.322	[-22.647, 4.883]	0.108	-16.643 [-45.293, 9.766]
Direct 1-1 animate patient vs inverse 2-2 animate patient	patient looks	Time 3 x contrast	2.836	[-11.223, 18.113]	0.650	5.671 [-22.446, 36.227]
Direct 1-1 inanimate patient vs inverse 2-2 inanimate patient	agent looks	Main contrast	0.060	[-0.464, 0.590]	0.597	0.120 [-0.928, 1.181]
Direct 1-1 inanimate patient vs inverse 2-2 inanimate patient	agent looks	Time 1 x contrast	-19.554	[-39.595, -0.347]	0.024	-39.108 [-79.190, -0.693]
Direct 1-1 inanimate patient vs inverse 2-2 inanimate patient	agent looks	Time 2 x contrast	-1.680	[-20.421, 17.662]	0.427	-3.361 [-40.842, 35.323]
Direct 1-1 inanimate patient vs inverse 2-2 inanimate patient	agent looks	Time 3 x contrast	-5.565	[-14.969, 3.694]	0.124	-11.130 [-29.938, 7.389]
Direct 1-1 inanimate patient vs inverse 2-2 inanimate patient	patient looks	Main contrast	0.060	[-0.543, 0.650]	0.586	0.120 [-1.087, 1.299]
Direct 1-1 inanimate patient vs inverse 2-2 inanimate patient	patient looks	Time 1 x contrast	13.251	[-8.983, 35.691]	0.890	26.501 [-17.966, 71.383]
Direct 1-1 inanimate patient vs inverse 2-2 inanimate patient	patient looks	Time 2 x contrast	0.147	[-21.064, 20.134]	0.515	0.295 [-42.129, 40.268]
Direct 1-1 inanimate patient vs inverse 2-2 inanimate patient	patient looks	Time 3 x contrast	-1.658	[-15.276, 10.968]	0.398	-3.316 [-30.553, 21.935]

Linguistic encoding 3

Comparison	Looks	Term	Beta	95% CI	P(beta > 0)	Full difference
Direct 1-1 animate patient vs direct 1-2 animate patient	agent looks	Main contrast	-0.113	[-0.669, 0.430]	0.326	-0.227 [-1.338, 0.860]
Direct 1-1 animate patient vs direct 1-2 animate patient	agent looks	Time 1 x contrast	-1.092	[-22.184, 19.977]	0.463	-2.185 [-44.367, 39.953]
Direct 1-1 animate patient vs direct 1-2 animate patient	agent looks	Time 2 x contrast	1.424	[-11.035, 12.699]	0.619	2.847 [-22.071, 25.398]
Direct 1-1 animate patient vs direct 1-2 animate patient	agent looks	Time 3 x contrast	-1.576	[-9.621, 7.054]	0.342	-3.152 [-19.242, 14.108]
Direct 1-1 animate patient vs direct 1-2 animate patient	patient looks	Main contrast	-0.096	[-0.679, 0.472]	0.375	-0.192 [-1.359, 0.945]
Direct 1-1 animate patient vs direct 1-2 animate patient	patient looks	Time 1 x contrast	-6.307	[-25.567, 13.304]	0.253	-12.613 [-51.133, 26.607]
Direct 1-1 animate patient vs direct 1-2 animate patient	patient looks	Time 2 x contrast	-7.869	[-18.645, 2.925]	0.071	-15.738 [-37.291, 5.849]
Direct 1-1 animate patient vs direct 1-2 animate patient	patient looks	Time 3 x contrast	3.770	[-3.524, 10.988]	0.857	7.541 [-7.049, 21.976]
Direct 1-1 inanimate patient vs direct 1-2 inanimate patient	agent looks	Main contrast	-0.174	[-0.565, 0.204]	0.189	-0.349 [-1.130, 0.409]
Direct 1-1 inanimate patient vs direct 1-2 inanimate patient	agent looks	Time 1 x contrast	10.915	[-13.192, 33.703]	0.826	21.830 [-26.384, 67.406]
Direct 1-1 inanimate patient vs direct 1-2 inanimate patient	agent looks	Time 2 x contrast	-13.442	[-30.559, 3.190]	0.057	-26.884 [-61.117, 6.379]
Direct 1-1 inanimate patient vs direct 1-2 inanimate patient	agent looks	Time 3 x contrast	-5.652	[-15.679, 4.070]	0.118	-11.303 [-31.357, 8.141]
Direct 1-1 inanimate patient vs direct 1-2 inanimate patient	patient looks	Main contrast	0.067	[-0.456, 0.551]	0.628	0.134 [-0.912, 1.102]
Direct 1-1 inanimate patient vs direct 1-2 inanimate patient	patient looks	Time 1 x contrast	-6.940	[-25.591, 10.261]	0.229	-13.879 [-51.182, 20.523]
Direct 1-1 inanimate patient vs direct 1-2 inanimate patient	patient looks	Time 2 x contrast	1.155	[-12.668, 15.720]	0.560	2.309 [-25.336, 31.439]
Direct 1-1 inanimate patient vs direct 1-2 inanimate patient	patient looks	Time 3 x contrast	7.754	[-3.729, 19.580]	0.908	15.508 [-7.458, 39.160]
Inverse 2-2 animate patient vs inverse 2-1 animate patient	agent looks	Main contrast	-0.179	[-0.726, 0.369]	0.248	-0.358 [-1.452, 0.738]
Inverse 2-2 animate patient vs inverse 2-1 animate patient	agent looks	Time 1 x contrast	-5.846	[-28.115, 17.517]	0.294	-11.691 [-56.230, 35.035]
Inverse 2-2 animate patient vs inverse 2-1 animate patient	agent looks	Time 2 x contrast	2.968	[-10.618, 16.948]	0.670	5.937 [-21.236, 33.896]
Inverse 2-2 animate patient vs inverse 2-1 animate patient	agent looks	Time 3 x contrast	-1.086	[-11.949, 9.654]	0.422	-2.173 [-23.899, 19.308]
Inverse 2-2 animate patient vs inverse 2-1 animate patient	patient looks	Main contrast	0.338	[-0.391, 1.059]	0.816	0.676 [-0.781, 2.117]

Comparison	Looks	Term	Beta	95% CI	P(beta > 0)	Full difference
Inverse 2-2 animate patient vs inverse 2-1 animate patient	patient looks	Time 1 x contrast	2.470	[-17.695, 22.868]	0.594	4.940 [-35.391, 45.735]
Inverse 2-2 animate patient vs inverse 2-1 animate patient	patient looks	Time 2 x contrast	2.465	[-6.521, 10.746]	0.734	4.931 [-13.042, 21.492]
Inverse 2-2 animate patient vs inverse 2-1 animate patient	patient looks	Time 3 x contrast	-2.371	[-9.539, 4.959]	0.259	-4.741 [-19.079, 9.919]
Direct 1-1 animate patient vs inverse 2-2 animate patient	agent looks	Main contrast	-0.405	[-1.111, 0.282]	0.126	-0.810 [-2.222, 0.563]
Direct 1-1 animate patient vs inverse 2-2 animate patient	agent looks	Time 1 x contrast	-0.459	[-19.176, 18.950]	0.476	-0.918 [-38.352, 37.901]
Direct 1-1 animate patient vs inverse 2-2 animate patient	agent looks	Time 2 x contrast	0.359	[-12.431, 13.060]	0.525	0.718 [-24.862, 26.120]
Direct 1-1 animate patient vs inverse 2-2 animate patient	agent looks	Time 3 x contrast	-1.679	[-9.960, 6.383]	0.337	-3.358 [-19.919, 12.766]
Direct 1-1 animate patient vs inverse 2-2 animate patient	patient looks	Main contrast	0.299	[-0.353, 0.915]	0.828	0.598 [-0.706, 1.829]
Direct 1-1 animate patient vs inverse 2-2 animate patient	patient looks	Time 1 x contrast	-4.443	[-19.191, 9.499]	0.273	-8.886 [-38.382, 18.998]
Direct 1-1 animate patient vs inverse 2-2 animate patient	patient looks	Time 2 x contrast	-5.956	[-17.956, 6.011]	0.159	-11.912 [-35.912, 12.023]
Direct 1-1 animate patient vs inverse 2-2 animate patient	patient looks	Time 3 x contrast	1.452	[-5.845, 8.896]	0.658	2.904 [-11.691, 17.791]

Fixed Effects: Intercept and Time Terms

Linguistic encoding 1

Comparison	Looks	Term	Beta	95% CI	P(beta > 0)
Direct 1-1 animate patient vs direct 1-2 animate patient	agent looks	Intercept	0.817	[0.400, 1.266]	1.000
Direct 1-1 animate patient vs direct 1-2 animate patient	agent looks	Time 1	20.157	[1.020, 40.426]	0.980
Direct 1-1 animate patient vs direct 1-2 animate patient	agent looks	Time 2	11.583	[-0.955, 24.408]	0.966
Direct 1-1 animate patient vs direct 1-2 animate patient	agent looks	Time 3	3.037	[-6.456, 12.255]	0.756
Direct 1-1 animate patient vs direct 1-2 animate patient	patient looks	Intercept	-2.250	[-3.049, -1.493]	0.000
Direct 1-1 animate patient vs direct 1-2 animate patient	patient looks	Time 1	-34.709	[-60.611, -8.547]	0.004
Direct 1-1 animate patient vs direct 1-2 animate patient	patient looks	Time 2	0.139	[-17.422, 18.975]	0.502
Direct 1-1 animate patient vs direct 1-2 animate patient	patient looks	Time 3	3.069	[-10.828, 17.573]	0.671
Direct 1-1 inanimate patient vs direct 1-2 inanimate patient	agent looks	Intercept	0.974	[0.305, 1.658]	0.995
Direct 1-1 inanimate patient vs direct 1-2 inanimate patient	agent looks	Time 1	29.849	[10.918, 49.654]	0.999
Direct 1-1 inanimate patient vs direct 1-2 inanimate patient	agent looks	Time 2	-0.817	[-17.816, 15.587]	0.461
Direct 1-1 inanimate patient vs direct 1-2 inanimate patient	agent looks	Time 3	-10.976	[-24.176, 0.606]	0.033
Direct 1-1 inanimate patient vs direct 1-2 inanimate patient	patient looks	Intercept	-2.101	[-2.888, -1.333]	0.000
Direct 1-1 inanimate patient vs direct 1-2 inanimate patient	patient looks	Time 1	-30.545	[-59.354, -1.584]	0.019
Direct 1-1 inanimate patient vs direct 1-2 inanimate patient	patient looks	Time 2	-1.021	[-20.562, 17.756]	0.466
Direct 1-1 inanimate patient vs direct 1-2 inanimate patient	patient looks	Time 3	5.746	[-6.474, 19.482]	0.813
Inverse 2-2 animate patient vs inverse 2-1 animate patient	agent looks	Intercept	0.804	[0.334, 1.294]	0.999
Inverse 2-2 animate patient vs inverse 2-1 animate patient	agent looks	Time 1	-8.661	[-31.773, 14.562]	0.232
Inverse 2-2 animate patient vs inverse 2-1 animate patient	agent looks	Time 2	3.611	[-22.983, 30.299]	0.614
Inverse 2-2 animate patient vs inverse 2-1 animate patient	agent looks	Time 3	11.495	[-0.091, 23.648]	0.973
Inverse 2-2 animate patient vs inverse 2-1 animate patient	patient looks	Intercept	-1.920	[-2.639, -1.211]	0.000
Inverse 2-2 animate patient vs inverse 2-1 animate patient	patient looks	Time 1	-6.802	[-35.543, 21.566]	0.309
Inverse 2-2 animate patient vs inverse 2-1 animate patient	patient looks	Time 2	6.761	[-15.938, 29.105]	0.732
Inverse 2-2 animate patient vs inverse 2-1 animate patient	patient looks	Time 3	-14.325	[-28.484, 0.565]	0.030
Direct 1-1 animate patient vs inverse 2-2 animate patient	agent looks	Intercept	0.813	[0.194, 1.479]	0.992

Comparison	Looks	Term	Beta	95% CI	P(beta > 0)
Direct 1-1 animate patient vs inverse 2-2 animate patient	agent looks	Time 1	11.586	[-13.335, 37.175]	0.815
Direct 1-1 animate patient vs inverse 2-2 animate patient	agent looks	Time 2	5.392	[-18.684, 30.180]	0.676
Direct 1-1 animate patient vs inverse 2-2 animate patient	agent looks	Time 3	11.060	[-0.808, 23.078]	0.966
Direct 1-1 animate patient vs inverse 2-2 animate patient	patient looks	Intercept	-2.260	[-3.034, -1.494]	0.000
Direct 1-1 animate patient vs inverse 2-2 animate patient	patient looks	Time 1	-27.925	[-55.705, -1.384]	0.018
Direct 1-1 animate patient vs inverse 2-2 animate patient	patient looks	Time 2	20.583	[-2.820, 44.348]	0.961
Direct 1-1 animate patient vs inverse 2-2 animate patient	patient looks	Time 3	-8.976	[-24.193, 6.785]	0.122
Direct 1-1 inanimate patient vs inverse 2-2 inanimate patient	agent looks	Intercept	1.254	[0.685, 1.913]	1.000
Direct 1-1 inanimate patient vs inverse 2-2 inanimate patient	agent looks	Time 1	43.588	[19.528, 68.170]	1.000
Direct 1-1 inanimate patient vs inverse 2-2 inanimate patient	agent looks	Time 2	-2.921	[-23.298, 17.164]	0.387
Direct 1-1 inanimate patient vs inverse 2-2 inanimate patient	agent looks	Time 3	-9.334	[-19.430, 0.420]	0.028
Direct 1-1 inanimate patient vs inverse 2-2 inanimate patient	patient looks	Intercept	-2.471	[-3.242, -1.731]	0.000
Direct 1-1 inanimate patient vs inverse 2-2 inanimate patient	patient looks	Time 1	-39.747	[-70.079, -9.984]	0.007
Direct 1-1 inanimate patient vs inverse 2-2 inanimate patient	patient looks	Time 2	8.291	[-15.815, 33.011]	0.750
Direct 1-1 inanimate patient vs inverse 2-2 inanimate patient	patient looks	Time 3	3.326	[-8.865, 15.807]	0.700

Linguistic encoding 3

Comparison	Looks	Term	Beta	95% CI	P(beta > 0)
Direct 1-1 animate patient vs direct 1-2 animate patient	agent looks	Intercept	-0.109	[-1.014, 0.805]	0.410
Direct 1-1 animate patient vs direct 1-2 animate patient	agent looks	Time 1	-28.968	[-49.924, -7.789]	0.006
Direct 1-1 animate patient vs direct 1-2 animate patient	agent looks	Time 2	-2.347	[-14.825, 9.742]	0.350
Direct 1-1 animate patient vs direct 1-2 animate patient	agent looks	Time 3	-3.575	[-10.941, 4.319]	0.165
Direct 1-1 animate patient vs direct 1-2 animate patient	patient looks	Intercept	-1.041	[-2.010, -0.125]	0.012
Direct 1-1 animate patient vs direct 1-2 animate patient	patient looks	Time 1	28.193	[10.469, 47.049]	0.998
Direct 1-1 animate patient vs direct 1-2 animate patient	patient looks	Time 2	-7.966	[-20.646, 3.563]	0.085
Direct 1-1 animate patient vs direct 1-2 animate patient	patient looks	Time 3	2.122	[-8.788, 13.312]	0.651
Direct 1-1 inanimate patient vs direct 1-2 inanimate patient	agent looks	Intercept	-1.011	[-2.116, 0.050]	0.030
Direct 1-1 inanimate patient vs direct 1-2 inanimate patient	agent looks	Time 1	-21.931	[-42.975, -0.579]	0.022
Direct 1-1 inanimate patient vs direct 1-2 inanimate patient	agent looks	Time 2	4.630	[-11.115, 21.449]	0.712
Direct 1-1 inanimate patient vs direct 1-2 inanimate patient	agent looks	Time 3	-6.140	[-17.021, 4.259]	0.111
Direct 1-1 inanimate patient vs direct 1-2 inanimate patient	patient looks	Intercept	-0.208	[-1.023, 0.634]	0.298
Direct 1-1 inanimate patient vs direct 1-2 inanimate patient	patient looks	Time 1	19.345	[2.363, 37.199]	0.984
Direct 1-1 inanimate patient vs direct 1-2 inanimate patient	patient looks	Time 2	-5.228	[-17.134, 6.366]	0.182
Direct 1-1 inanimate patient vs direct 1-2 inanimate patient	patient looks	Time 3	18.222	[9.327, 26.742]	1.000
Inverse 2-2 animate patient vs inverse 2-1 animate patient	agent looks	Intercept	0.893	[0.040, 1.779]	0.980
Inverse 2-2 animate patient vs inverse 2-1 animate patient	agent looks	Time 1	-24.220	[-44.544, -3.791]	0.011
Inverse 2-2 animate patient vs inverse 2-1 animate patient	agent looks	Time 2	-1.930	[-16.237, 11.802]	0.402
Inverse 2-2 animate patient vs inverse 2-1 animate patient	agent looks	Time 3	0.132	[-8.021, 8.452]	0.516
Inverse 2-2 animate patient vs inverse 2-1 animate patient	patient looks	Intercept	-2.000	[-2.821, -1.191]	0.000
Inverse 2-2 animate patient vs inverse 2-1 animate patient	patient looks	Time 1	29.065	[11.595, 45.818]	0.999
Inverse 2-2 animate patient vs inverse 2-1 animate patient	patient looks	Time 2	-5.174	[-20.721, 10.618]	0.249
Inverse 2-2 animate patient vs inverse 2-1 animate patient	patient looks	Time 3	6.755	[-1.562, 14.672]	0.948

Comparison	Looks	Term	Beta	95% CI	P(beta > 0)
Direct 1-1 animate patient vs inverse 2-2 animate patient	agent looks	Intercept	0.150	[-0.706, 0.973]	0.646
Direct 1-1 animate patient vs inverse 2-2 animate patient	agent looks	Time 1	-30.920	[-54.700, -8.087]	0.006
Direct 1-1 animate patient vs inverse 2-2 animate patient	agent looks	Time 2	-1.086	[-13.940, 11.545]	0.434
Direct 1-1 animate patient vs inverse 2-2 animate patient	agent looks	Time 3	-2.128	[-12.067, 8.055]	0.331
Direct 1-1 animate patient vs inverse 2-2 animate patient	patient looks	Intercept	-1.427	[-2.412, -0.464]	0.002
Direct 1-1 animate patient vs inverse 2-2 animate patient	patient looks	Time 1	27.110	[10.027, 44.848]	0.999
Direct 1-1 animate patient vs inverse 2-2 animate patient	patient looks	Time 2	-7.550	[-14.395, -1.150]	0.011
Direct 1-1 animate patient vs inverse 2-2 animate patient	patient looks	Time 3	5.277	[-3.174, 13.668]	0.896

Convergence Diagnostics

Fixed-effect diagnostics are broadly acceptable for completed models. All-parameter diagnostics are stricter because they include nuisance random-effect SD/correlation parameters.

Fixed Effects

Window	Comparison	Looks	Max Rhat	Min bulk ESS	Min tail ESS
Linguistic encoding 1	Direct 1-1 animate patient vs direct 1-2 animate patient	agent looks	1.0049	1778	2183
Linguistic encoding 1	Direct 1-1 animate patient vs direct 1-2 animate patient	patient looks	1.0047	1224	1429
Linguistic encoding 1	Direct 1-1 inanimate patient vs direct 1-2 inanimate patient	agent looks	1.0031	1160	1770
Linguistic encoding 1	Direct 1-1 inanimate patient vs direct 1-2 inanimate patient	patient looks	1.0058	1105	1664
Linguistic encoding 1	Inverse 2-2 animate patient vs inverse 2-1 animate patient	agent looks	1.0078	1876	2120
Linguistic encoding 1	Inverse 2-2 animate patient vs inverse 2-1 animate patient	patient looks	1.0064	1385	1976
Linguistic encoding 1	Direct 1-1 animate patient vs inverse 2-2 animate patient	agent looks	1.0067	822	1414
Linguistic encoding 1	Direct 1-1 animate patient vs inverse 2-2 animate patient	patient looks	1.0053	956	1300
Linguistic encoding 1	Direct 1-1 inanimate patient vs inverse 2-2 inanimate patient	agent looks	1.0025	1423	2129
Linguistic encoding 1	Direct 1-1 inanimate patient vs inverse 2-2 inanimate patient	patient looks	1.0052	1287	1844
Linguistic encoding 3	Direct 1-1 animate patient vs direct 1-2 animate patient	agent looks	1.0086	981	1659
Linguistic encoding 3	Direct 1-1 animate patient vs direct 1-2 animate patient	patient looks	1.0061	1130	1935
Linguistic encoding 3	Direct 1-1 inanimate patient vs direct 1-2 inanimate patient	agent looks	1.0097	624	1067
Linguistic encoding 3	Direct 1-1 inanimate patient vs direct 1-2 inanimate patient	patient looks	1.0038	623	1282
Linguistic encoding 3	Inverse 2-2 animate patient vs inverse 2-1 animate patient	agent looks	1.0051	1352	1818
Linguistic encoding 3	Inverse 2-2 animate patient vs inverse 2-1 animate patient	patient looks	1.0047	1656	2319
Linguistic encoding 3	Direct 1-1 animate patient vs inverse 2-2 animate patient	agent looks	1.0031	1587	2032
Linguistic encoding 3	Direct 1-1 animate patient vs inverse 2-2 animate patient	patient looks	1.0026	1004	1783

All Parameters

Window	Comparison	Looks	Max Rhat	Min bulk ESS	Min tail ESS	Worst bulk variable	Worst tail variable
Linguistic encoding 1	Direct 1-1 animate patient vs direct 1-2 animate patient	agent looks	1.2462	12	45	sd_participant_condition__polytimedegreeEQ31	sd_participant__polytimedegreeEQ31
Linguistic encoding 1	Direct 1-1 animate patient vs direct 1-2 animate patient	patient looks	1.0197	389	273	sd_participant_contrast_code	sd_participant_condition__polytimedegreeEQ33
Linguistic encoding 1	Direct 1-1 inanimate patient vs direct 1-2 inanimate patient	agent looks	1.0574	47	25	sd_participant_condition__polytimedegreeEQ32	r_participant_condition[K24.direct_1_agent_1_patient_inanimate_patient,polytimedegreeEQ32]

Window	Comparison	Looks	Max Rhat	Min bulk ESS	Min tail ESS	Worst bulk variable	Worst tail variable
Linguistic encoding 1	Direct 1-1 inanimate patient vs direct 1-2 inanimate patient	patient looks	1.0339	84	22	r_participant_condition[K15.direct_1_agent_1_patient_inanimate_patient,polytimedegreeEQ32]	r_participant_condition[K13.direct_1_agent_1_patient_inanimate_patient,polytimedegreeEQ32]
Linguistic encoding 1	Inverse 2-2 animate patient vs inverse 2-1 animate patient	agent looks	1.7343	6	102	sd_participant_condition__polytimedegreeEQ32	r_participant_condition[K10.inverse_2_agents_2_patients_animate_patient,polytimedegreeEQ32]
Linguistic encoding 1	Inverse 2-2 animate patient vs inverse 2-1 animate patient	patient looks	1.0278	198	199	cor_participant_condition__Intercept__polytimedegreeEQ31	cor_participant_condition__Intercept__polytimedegreeEQ31
Linguistic encoding 1	Direct 1-1 animate patient vs inverse 2-2 animate patient	agent looks	1.4598	8	104	sd_participant_condition__polytimedegreeEQ32	r_participant_condition[K06.inverse_2_agents_2_patients_animate_patient,polytimedegreeEQ32]
Linguistic encoding 1	Direct 1-1 animate patient vs inverse 2-2 animate patient	patient looks	1.0239	118	225	r_participant_condition[K24.direct_1_agent_1_patient_animate_patient,polytimedegreeEQ32]	cor_participant_condition__Intercept__polytimedegreeEQ31
Linguistic encoding 1	Direct 1-1 inanimate patient vs inverse 2-2 inanimate patient	agent looks	1.0248	153	34	r_participant_condition[K17.inverse_2_agents_2_patients_inanimate_patient,polytimedegreeEQ31]	sd_participant_condition__polytimedegreeEQ31
Linguistic encoding 1	Direct 1-1 inanimate patient vs inverse 2-2 inanimate patient	patient looks	1.1136	26	64	r_participant_condition[K20.direct_1_agent_1_patient_inanimate_patient,polytimedegreeEQ31]	sd_participant_condition__polytimedegreeEQ32
Linguistic encoding 3	Direct 1-1 animate patient vs direct 1-2 animate patient	agent looks	1.0482	57	22	sd_participant__polytimedegreeEQ31	r_participant[K16,polytimedegreeEQ31]
Linguistic encoding 3	Direct 1-1 animate patient vs direct 1-2 animate patient	patient looks	1.0121	291	283	sd_participant_condition__polytimedegreeEQ32	sd_participant__Intercept
Linguistic encoding 3	Direct 1-1 inanimate patient vs direct 1-2 inanimate patient	agent looks	1.0132	485	695	cor_participant_condition__Intercept__polytimedegreeEQ31	sd_item__polytimedegreeEQ33
Linguistic encoding 3	Direct 1-1 inanimate patient vs direct 1-2 inanimate patient	patient looks	1.0989	28	62	sd_participant_condition__polytimedegreeEQ31	r_participant_condition[K16.direct_1_agent_2_patients_inanimate_patient,polytimedegreeEQ31]
Linguistic encoding 3	Inverse 2-2 animate patient vs inverse 2-1 animate patient	agent looks	1.0126	483	727	cor_participant_condition__Intercept__polytimedegreeEQ31	cor_participant_condition__Intercept__polytimedegreeEQ31
Linguistic encoding 3	Inverse 2-2 animate patient vs inverse 2-1 animate patient	patient looks	1.0307	114	469	sd_participant_condition__polytimedegreeEQ31	r_participant_condition[K24.inverse_2_agents_2_patients_animate_patient,polytimedegreeEQ31]
Linguistic encoding 3	Direct 1-1 animate patient vs inverse 2-2 animate patient	agent looks	1.0068	429	240	sd_participant_condition__Intercept	r_participant_condition[K17.inverse_2_agents_2_patients_animate_patient,Intercept]
Linguistic encoding 3	Direct 1-1 animate patient vs inverse 2-2 animate patient	patient looks	1.0339	163	116	sd_participant_condition__Intercept	sd_participant_condition__polytimedegreeEQ31

Sampler Diagnostics

Window	Comparison	Looks	Divergences	Max-treepdepth hits
Linguistic encoding 1	Direct 1-1 animate patient vs direct 1-2 animate patient	agent looks	0	0
Linguistic encoding 1	Direct 1-1 animate patient vs direct 1-2 animate patient	patient looks	0	0
Linguistic encoding 1	Direct 1-1 inanimate patient vs direct 1-2 inanimate patient	agent looks	0	0
Linguistic encoding 1	Direct 1-1 inanimate patient vs direct 1-2 inanimate patient	patient looks	0	0
Linguistic encoding 1	Inverse 2-2 animate patient vs inverse 2-1 animate patient	agent looks	0	0
Linguistic encoding 1	Inverse 2-2 animate patient vs inverse 2-1 animate patient	patient looks	0	0
Linguistic encoding 1	Direct 1-1 animate patient vs inverse 2-2 animate patient	agent looks	0	0
Linguistic encoding 1	Direct 1-1 animate patient vs inverse 2-2 animate patient	patient looks	0	0
Linguistic encoding 1	Direct 1-1 inanimate patient vs inverse 2-2 inanimate patient	agent looks	0	0
Linguistic encoding 1	Direct 1-1 inanimate patient vs inverse 2-2 inanimate patient	patient looks	0	0
Linguistic encoding 3	Direct 1-1 animate patient vs direct 1-2 animate patient	agent looks	0	0
Linguistic encoding 3	Direct 1-1 animate patient vs direct 1-2 animate patient	patient looks	0	0
Linguistic encoding 3	Direct 1-1 inanimate patient vs direct 1-2 inanimate patient	agent looks	0	0
Linguistic encoding 3	Direct 1-1 inanimate patient vs direct 1-2 inanimate patient	patient looks	0	0
Linguistic encoding 3	Inverse 2-2 animate patient vs inverse 2-1 animate patient	agent looks	0	0
Linguistic encoding 3	Inverse 2-2 animate patient vs inverse 2-1 animate patient	patient looks	0	0

Window	Comparison	Looks	Divergences	Max-treedepth hits
Linguistic encoding 3	Direct 1-1 animate patient vs inverse 2-2 animate patient	agent looks	0	0
Linguistic encoding 3	Direct 1-1 animate patient vs inverse 2-2 animate patient	patient looks	0	0